

UNIVERSITI TUNKU ABDUL RAHMAN

ACADEMIC YEAR 2023/2024

MAY EXAMINATION

UCCC3073 / UCCG3073 DATA SCIENCE

TUESDAY, 14 MAY 2024

TIME : 9.00 AM – 11.00 AM (2 HOURS)

BACHELOR OF COMPUTER SCIENCE (HONOURS)
BACHELOR OF SCIENCE (HONOURS) STATISTICAL COMPUTING AND
OPERATIONAL RESEARCH

Instruction to Candidates:

This question paper consists of **FIVE (5)** questions.

Answer **ALL** questions in **Section A** and **ONLY ONE (1)** question in **Section B**. Each question carries 25 marks.

Should a candidate answers more than ONE (1) question in Section B, marks will only be awarded for the **FIRST (1)** question in that section in the order the candidate submits the answers.

Candidates are allowed to use any type of scientific calculator.

Answer questions only in the answer booklet provided.

UCCC3073 / UCCG3073 DATA SCIENCESECTION A (Answer ALL questions)

Q1. (a) AquaScratch Ltd. specializes in the installation of wall-mounted aquariums for its clientele. Each customer may opt for multiple installations. Various fittings, including thermostats, oxygen pumps, and aquariums, are utilized in these installations. It is possible for a specific fitting to be employed across multiple installations. Furthermore, it is imperative to keep track of the quantity of fittings used in each installation.

- (i) Identify **THREE (3)** entities for data modeling. (3 marks)
- (ii) Using the answers of Q1(a)(i), sketch a logical model to illustrate relationships between entities. In the model, provide at least **TWO (2)** attributes for each entity and identify keys(s). (8 marks)
- (iii) Suggest appropriate data types and constraints for each attribute outlined in Q1(a)(ii). (6 marks)

(b) Consider the following linear scaled data:

0.66, 0.28, 1.65, 1.81, 1.41, 0.25, 0, 2

- (i) Determine the range of the data. (1 mark)
- (ii) Given that the minimum and the maximum values of the original data are -67 and 69, inverse transform the data back to its original value. Round your answers to the nearest integer. (7 marks)

[Total : 25 marks]

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- Q2. (a) A team of scientists test a new medication to see if it has effect on human IQ. The population average IQ is 100 with a standard deviation of 40. A sample of 50 participants were recruited to take the medication and has a mean of 90. A hypothesis testing is carried out for the following:

H_0 = New medication does not have effect on human IQ.

H_a = New medication has effect on human IQ.

Conduct a hypothesis test to ascertain whether there is substantial evidence to uphold the claim at a 5% significance level. Determine the following:

- (i) z test. Round your answer to 2 decimal points. (3 marks)
 - (ii) p-value. Round your answer to 4 decimal points. (2 marks)
 - (iii) upper- and lower-bounds. Round your answer to the nearest integer. (4 marks)
- (b) Using the answers of Q2(a), justify whether the null hypothesis can be rejected. (3 marks)
- (c) Describe **TWO (2)** types of hypothesis errors can be made. For each type of errors, suggest a way to reduce the error in the hypothesis testing. (6 marks)
- (d) Given a bag of 20 balls of which 6 are red and the remaining are blue. Two (2) balls were drawn from the bag and did not put back. Draw the probability tree to determine the probability that the drew balls are different in color. (7 marks)
- [Total : 25 marks]

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- Q3. (a) Given the following dataset of duration of sunshine (in hour) and amount of rainfall (in mm) in Kampar for the second half of 2023.

Table 3.1

Month	Sunshine (hours)	Rainfall (mm)
July	294	128
August	293	152
September	264	275
October	262	378
November	227	433
December	270	311

- (i) Determine the Pearson's correlation coefficient, r , between duration of sunshine (in hour) and amount of rainfall (in mm). (8 marks)
- (ii) Using the answer of Q3(a)(i), describe the relationship between duration of sunshine and amount of rainfall. (4 marks)
- (b) Briefly describe the association rules mining technique. (3 marks)
- (c) Consider the following transactions:

Table 3.2

Transaction ID	Bread	Cabbage	Durian	Milk	Peanut
T1	1	0	0	1	1
T2	1	0	1	1	1
T3	1	1	1	0	1
T4	1	0	1	0	1
T5	1	0	1	1	0
T6	1	0	0	0	1
T7	0	0	0	1	1

Determine support, confidence and lift for the following association rule:

$\{\text{Bread, Durian}\} \rightarrow \{\text{Peanut}\}$

(10 marks)

[Total : 25 marks]

UCCC3073 / UCCG3073 DATA SCIENCE**SECTION B (Choose only ONE question)**

- Q4. Suppose you have developed a classification model to predict whether emails are spam or not spam. Table 4.1 is the confusion matrix made by the model on the test data:

Table 4.1

		Prediction	
		Not spam	Spam
Target/ Actual	Not spam	9000	100
	Spam	400	500

- (a) Determine the total number of samples in the data. (1 mark)
- (b) Identify positive and negative classes in the data. (2 marks)
- (c) Calculate the baseline performance. (2 marks)
- (d) Calculate the following evaluation measures. Round your answer to the nearest 2 decimal points.
- (i) Misclassification rate. (2 marks)
- (ii) Precision. (2 marks)
- (iii) Recall. (2 marks)
- (iv) F_1 score. (2 marks)
- (e) Using the computed results in Q4(c) and Q4(d), justify whether you would propose the model to the company. (7 marks)
- (f) Suggest **TWO (2)** ways to overcome issues you have observed in Q4(e). (5 marks)
- [Total : 25 marks]

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- Q5. You are a data scientist working in a company. You are tasked with developing a machine learning model to classify in-coming emails into two categories: "Not spam" and "Spam". The company has provided a breakdown of the costs associated with misclassification. A false positive (FP) incurs a cost of RM500 due to additional retrieval process, while a false negative (FN) incurs a significantly higher cost of RM1,500 due to potential complications. Table 5.1 shows the prediction results of the models on the test data.

Table 5.1

		Prediction	
		Not spam	Spam
Target/ Actual	Not spam	4800	100
	Spam	460	640

- (a) Determine the total number of samples in the data. (1 mark)
- (b) Identify positive and negative classes in the data. (2 marks)
- (c) Calculate the baseline performance. Round your answer to the nearest 2 decimal points. (2 marks)
- (d) Calculate the following evaluation measures. Round your answer to the nearest 2 decimal points.
- (i) Classification accuracy of the model. (2 marks)
- (ii) Overall model cost and model profit per record. (6 marks)
- (e) Using the computed results in Q5(c) and Q5(d), justify whether you would propose the model to the company. (5 marks)
- (f) Assume that the company has revised the costs associated with misclassification. A new misclassification cost for FP is remain unchanged, and a FN cost was reduced to RM750. Calculate the model profit per record and justify whether you would still propose the model to the company. (7 marks)

[Total : 25 marks]

APPENDIX

Proximity measures	
Euclidean distance $d(x, y) = \sqrt{\sum_{i=1}^n (x_i - x)^2 + (y_i - y)^2}$ Manhattan distance $d(x, y) = \sum_{i=1}^n x_i - y_i $ $d(x, y) = \sum_{i=1}^n x_i - x + y_i - y $ Supremum distance $d(x, y) = \max(x_i - y_i)$ $d(x, y) = \max(x_i - x , y_i - y)$	Hamming distance $d(a, b) = (a \oplus b)$ Simple matching coefficient $SMC = \frac{\text{num of matching attribute values}}{\text{num of attributes}}$ Jaccard coefficient $J = \frac{\text{num of matching presences}}{\text{num of attributes not involved in 00 matches}}$ Cosine similarity $\cos(x, y) = \frac{x \cdot y}{ x y }$
Regression	
Straight line $\hat{y} = \beta_0 + \beta_1 x$ Coefficients $\hat{\beta}_1 = \frac{p \sum xy - \sum x \sum y}{p \sum x^2 - (\sum x)^2}$ $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$ Residuals $\hat{\varepsilon} = y - \hat{y}$ $s^2 = \frac{\sum \hat{\varepsilon}^2}{p - 2}$ $s = \sqrt{s^2}$ t value $t = \frac{\hat{\beta}}{SE}$	Coefficient of Determination $R^2 = \left[\frac{p \sum xy - \sum x \sum y}{\sqrt{(p \sum x^2 - (\sum x)^2) * (p \sum y^2 - (\sum y)^2)}} \right]^2$ Pearson correlation $R = \sqrt{R^2}$ Standard errors of coefficients $SE(\hat{\beta}_1) = \frac{s}{\sqrt{\sum (x - \bar{x})^2}}$ $SE(\hat{\beta}_0) = s \sqrt{\frac{1}{p} + \frac{\bar{x}^2}{\sum (x - \bar{x})^2}}$
Naïve Bayes	
Bayes' rule $P(H E) = \frac{P(E H)P(H)}{P(E)}$ Probability density function $pdf(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{\frac{-(x-\bar{x})^2}{2\sigma^2}}$	Laplace correction $\frac{x_i + 1}{c + \alpha}$

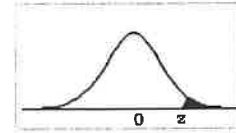
Association Rules	
Association rule $X \rightarrow Y$	Confidence $c = \frac{freq(X, Y)}{freq(X)}$
Support $s = \frac{freq(X, Y)}{N}$	Lift $l = \frac{Confidence(X, Y)}{Support(X)}$

Evaluation metrics	
Classification accuracy $acc = \frac{TN + TP}{TN + TP + FN + FP}$	Precision / Positive Predictive Value $PPV = \frac{TP}{TP + FP} = 1 - FDR$
Misclassification rate $1 - acc$	F1 score $F_1 = \frac{2TP}{2TP + FP + FN}$
Sensitivity / Recall $recall = \frac{TP}{TP + FN}$	Overall model cost $FP.Cost_{FP} + TP.Cost_{TP}$
Specificity $spec = \frac{TN}{TN + FP}$	Model cost per case $\frac{Overall\ model\ cost}{N}$
False discovery rate $FDR = \frac{FP}{FP + TP}$	Model profit per case $-model\ cost\ per\ case$

Miscellaneous	
Linear scaling for [a, b] $x' = (b - a) \frac{x - \min(x)}{\max(x) - \min(x)} + a$	Inverse linear scaling for [a, b] $x = (x' - a) * \frac{\max(x) - \min(x)}{b - a} + \min(x)$
z-score normalisation $x' = \frac{x - \mu}{\sigma}$	Population standard deviation $\sigma = \sqrt{\frac{\sum_{i=1}^n (x_i - \mu)^2}{n}}$
Mean, $\bar{x}$ $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$	Sample standard deviation $s = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1}}$
Z-test $Z = \frac{\bar{x} - \mu}{SE_{\bar{x}}}$	Standard error $SE = \frac{\sigma}{\sqrt{n}}$
t-test $t = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}}$	Confidence intervals $CI = \bar{x} \pm z * SE$

UCCC3073 / UCCG3073 DATA SCIENCE**Normal Curve Areas (Standard normal probability in right-hand tail)**for negative values of z , areas are found by symmetry confidence level to Z-score

90%	1.645
95%	1.96
98%	2.33
99%	2.575



z	Second decimal place of z									
	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.4960	0.4920	0.4880	0.4840	0.4801	0.4761	0.4721	0.4681	0.4641
0.1	0.4602	0.4562	0.4522	0.4483	0.4443	0.4404	0.4364	0.4325	0.4286	0.4247
0.2	0.4207	0.4168	0.4129	0.4090	0.4052	0.4013	0.3974	0.3936	0.3897	0.3859
0.3	0.3821	0.3783	0.3745	0.3707	0.3669	0.3632	0.3594	0.3557	0.3520	0.3483
0.4	0.3446	0.3409	0.3372	0.3336	0.3300	0.3264	0.3228	0.3192	0.3156	0.3121
0.5	0.3085	0.3050	0.3015	0.2981	0.2946	0.2912	0.2877	0.2843	0.2810	0.2776
0.6	0.2743	0.2709	0.2676	0.2643	0.2611	0.2578	0.2546	0.2514	0.2483	0.2451
0.7	0.2420	0.2389	0.2358	0.2327	0.2296	0.2266	0.2236	0.2206	0.2177	0.2148
0.8	0.2119	0.2090	0.2061	0.2033	0.2005	0.1977	0.1949	0.1922	0.1894	0.1867
0.9	0.1841	0.1814	0.1788	0.1762	0.1736	0.1711	0.1685	0.1660	0.1635	0.1611
1.0	0.1587	0.1562	0.1539	0.1515	0.1492	0.1469	0.1446	0.1423	0.1401	0.1379
1.1	0.1357	0.1335	0.1314	0.1292	0.1271	0.1251	0.1230	0.1210	0.1190	0.1170
1.2	0.1151	0.1131	0.1112	0.1093	0.1075	0.1056	0.1038	0.1020	0.1003	0.0985
1.3	0.0968	0.0951	0.0934	0.0918	0.0901	0.0885	0.0869	0.0853	0.0838	0.0823
1.4	0.0808	0.0793	0.0778	0.0764	0.0749	0.0735	0.0721	0.0708	0.0694	0.0681
1.5	0.0668	0.0655	0.0643	0.0630	0.0618	0.0606	0.0594	0.0582	0.0571	0.0559
1.6	0.0548	0.0537	0.0526	0.0516	0.0505	0.0495	0.0485	0.0475	0.0465	0.0455
1.7	0.0446	0.0436	0.0427	0.0418	0.0409	0.0401	0.0392	0.0384	0.0375	0.0367
1.8	0.0359	0.0351	0.0344	0.0336	0.0329	0.0322	0.0314	0.0307	0.0301	0.0294
1.9	0.0287	0.0281	0.0274	0.0268	0.0262	0.0256	0.0250	0.0244	0.0239	0.0233
2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183
2.1	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	0.0150	0.0146	0.0143
2.2	0.0139	0.0136	0.0132	0.0129	0.0125	0.0122	0.0119	0.0116	0.0113	0.0110
2.3	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084
2.4	0.0082	0.0080	0.0078	0.0075	0.0073	0.0071	0.0069	0.0068	0.0066	0.0064
2.5	0.0062	0.0060	0.0059	0.0057	0.0055	0.0054	0.0052	0.0051	0.0049	0.0048
2.6	0.0047	0.0045	0.0044	0.0043	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036
2.7	0.0035	0.0034	0.0033	0.0032	0.0031	0.0030	0.0029	0.0028	0.0027	0.0026
2.8	0.0026	0.0025	0.0024	0.0023	0.0023	0.0022	0.0021	0.0021	0.0020	0.0019
2.9	0.0019	0.0018	0.0018	0.0017	0.0016	0.0016	0.0015	0.0015	0.0014	0.0014
3.0	0.0013	0.0013	0.0013	0.0012	0.0012	0.0011	0.0011	0.0011	0.0010	0.0010
3.1	0.0010	0.0009	0.0009	0.0009	0.0008	0.0008	0.0008	0.0008	0.0007	0.0007
3.2	0.0007	0.0007	0.0006	0.0006	0.0006	0.0006	0.0006	0.0005	0.0005	0.0005
3.3	0.0005	0.0005	0.0005	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0003
3.4	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0002
3.5	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002
3.6	0.0002	0.0002	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001
3.7	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001
3.8	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001
3.9	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
4.0	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000

UCCC3073 / UCCG3073 DATA SCIENCE**t Distribution: Critical Values of t**

Degree of Freedom	1-tailed α	Significant level						
		0.25	0.2	0.15	0.1	0.05	0.025	0.01
2-tailed α		0.5	0.4	0.3	0.2	0.1	0.05	0.02
1		1.000	1.376	1.963	3.078	6.314	12.706	31.821
2		0.816	1.061	1.386	1.886	2.920	4.303	6.965
3		0.765	0.978	1.250	1.638	2.353	3.182	4.541
4		0.741	0.941	1.190	1.533	2.132	2.776	3.747
5		0.727	0.920	1.156	1.476	2.015	2.571	3.365
6		0.718	0.906	1.134	1.440	1.943	2.447	3.143
7		0.711	0.896	1.119	1.415	1.895	2.365	2.998
8		0.706	0.889	1.108	1.397	1.860	2.306	2.896
9		0.703	0.883	1.100	1.383	1.833	2.262	2.821
10		0.700	0.879	1.093	1.372	1.812	2.228	2.764
11		0.697	0.876	1.088	1.363	1.796	2.201	2.718
12		0.695	0.873	1.083	1.356	1.782	2.179	2.681
13		0.694	0.870	1.079	1.350	1.771	2.160	2.650
14		0.692	0.868	1.076	1.345	1.761	2.145	2.624
15		0.691	0.866	1.074	1.341	1.753	2.131	2.602
16		0.690	0.865	1.071	1.337	1.746	2.120	2.583
17		0.689	0.863	1.069	1.333	1.740	2.110	2.567
18		0.688	0.862	1.067	1.330	1.734	2.101	2.552
19		0.688	0.861	1.066	1.328	1.729	2.093	2.539
20		0.687	0.860	1.064	1.325	1.725	2.086	2.528
21		0.686	0.859	1.063	1.323	1.721	2.080	2.518
22		0.686	0.858	1.061	1.321	1.717	2.074	2.508
23		0.685	0.858	1.060	1.319	1.714	2.069	2.500
24		0.685	0.857	1.059	1.318	1.711	2.064	2.492
25		0.684	0.856	1.058	1.316	1.708	2.060	2.485
26		0.684	0.856	1.058	1.315	1.706	2.056	2.479
27		0.684	0.855	1.057	1.314	1.703	2.052	2.473
28		0.683	0.855	1.056	1.313	1.701	2.048	2.467
29		0.683	0.854	1.055	1.311	1.699	2.045	2.462
30		0.683	0.854	1.055	1.310	1.697	2.042	2.457
31		0.682	0.853	1.054	1.309	1.696	2.040	2.453
32		0.682	0.853	1.054	1.309	1.694	2.037	2.449
33		0.682	0.853	1.053	1.308	1.692	2.035	2.445
34		0.682	0.852	1.052	1.307	1.691	2.032	2.441
35		0.682	0.852	1.052	1.306	1.690	2.030	2.438
36		0.681	0.852	1.052	1.306	1.688	2.028	2.434
37		0.681	0.851	1.051	1.305	1.687	2.026	2.431
38		0.681	0.851	1.051	1.304	1.686	2.024	2.429
39		0.681	0.851	1.050	1.304	1.685	2.023	2.426
40		0.681	0.851	1.050	1.303	1.684	2.021	2.423
41		0.681	0.850	1.050	1.303	1.683	2.020	2.421
42		0.680	0.850	1.049	1.302	1.682	2.018	2.418
43		0.680	0.850	1.049	1.302	1.681	2.017	2.416
44		0.680	0.850	1.049	1.301	1.680	2.015	2.414
45		0.680	0.850	1.049	1.301	1.679	2.014	2.412
46		0.680	0.850	1.048	1.300	1.679	2.013	2.410
47		0.680	0.849	1.048	1.300	1.678	2.012	2.408
48		0.680	0.849	1.048	1.299	1.677	2.011	2.407
49		0.680	0.849	1.048	1.299	1.677	2.010	2.405
50		0.679	0.849	1.047	1.299	1.676	2.009	2.403
60		0.679	0.848	1.045	1.296	1.671	2.000	2.390
70		0.678	0.847	1.044	1.294	1.667	1.994	2.381
80		0.678	0.846	1.043	1.292	1.664	1.990	2.374
90		0.677	0.846	1.042	1.291	1.662	1.987	2.368
100		0.677	0.845	1.042	1.290	1.660	1.984	2.364
150		0.676	0.844	1.040	1.287	1.655	1.976	2.351
200		0.676	0.843	1.039	1.286	1.653	1.972	2.345
		0.675	0.842	1.036	1.282	1.645	1.960	2.327